

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 17

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• $(6400 - 220) / 6400 \times 100$ • = 96.6% (accept 97%)	2
01.2	<i>Any 3 from:</i> • Switch from coal to natural gas (which is desulfurised) / closure of coal-fired power stations removed the main SO₂ source • Flue gas desulfurisation fitted to remaining coal plants; coal / oil desulfurisation • NO_x comes mainly from road transport and vehicle numbers / distance travelled increased greatly • NO_x is produced from air (nitrogen + oxygen) in any high-temperature combustion, so cannot be removed from the fuel like sulfur • Catalytic converters only work when warm / diesel engines produce more NO_x / older vehicles remained in use	3
01.3	• Ozone and PANs are secondary pollutants formed by photochemical reactions involving NO₂ (broken down by UV to monatomic oxygen which forms ozone) • If the primary pollutant is reduced the reactants are not present so the secondary pollutants cannot form	2
01.4	• Lichens are very sensitive to acidic conditions / SO₂ and different species have different tolerances • The species present, their abundance, size / health can be used as a biotic index of the pollution level • They give a long-term integrated measure of pollution rather than a single instantaneous reading; cheap, no equipment needed	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-253; AQA spec 3.4.3.2.1-3.4.3.2.3.	

Question	Marking guidance	Marks
02.1	- The energy used to extract raw materials, manufacture the components and assemble the vehicle (for a typical car about 1.5 years of its fuel use) - Reduced by using recycled materials (eg recycled aluminium needs 1/20 of the energy of new) - Lighter designs using less material; longer vehicle life spreading the embodied energy	3
02.2	<i>Any 4 from:</i> - Use of recyclable materials wherever possible - Easy identification of components and their composition (code stamps) - Easy dismantling and separation of components / materials - Reusable components for new vehicles - Compostable materials for parts that cannot be recycled	4
02.3	- At low speed a large proportion of the energy just runs the engine rather than moving the car - At high speed energy is wasted overcoming air resistance (which rises with the square of speed); optimum for many cars about 56 mph	2
02.4	Automatic stop / start systems cutting the engine when stationary; smooth driving in higher gears avoiding sudden braking / acceleration	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 227-228; AQA spec 3.3.4.2.1.	
03.1	- Quadrat size depends on the size / spacing of the organisms: 0.5 m x 0.5 m is typical for ground flora 10 cm x 10 cm is for lichens (too small to contain a representative sample of plants); 100 m x 100 m is for trees (impractical, too much work)	2
03.2	$N = 240$; $N(N-1) = 240 \times 239 = 57\,360$ $\Sigma n(n-1) = 48 \times 47 + 120 \times 119 + 36 \times 35 + 24 \times 23 + 12 \times 11 = 2256 + 14280 + 1260 + 552 + 132 = 18\,480$ $D = 57\,360 / 18\,480 = 3.10$ (accept 3.1)	3
03.3	- Lower D means lower biodiversity: the ungrazed meadow is dominated by one or a few species (eg tall grass) with the others rare (less even) - Grazing (plagioclimax management) prevents dominance by vigorous species so abundance is more evenly spread	2
03.4	- A frame quadrat flattens tall plants, making percentage cover estimates inaccurate and hiding shorter plants - Point quadrat pins are lowered through the vegetation; each touch counts (10 pins = 10% each) without flattening	2
03.5	Subjective judgement varies between recorders; results are not quantitative so statistical analysis is not possible (but quick for many areas)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 394-395, 401-402; AQA spec 3.7.2.1.	
04.1	- Friction with the prevailing south-westerly winds drags surface water north-eastwards from the tropical Atlantic - Water in the NE Atlantic cools, becomes denser (and more saline as ice forms) and sinks, drawing warm surface water in to replace it; the cold water returns southwards at depth (thermohaline circulation)	2

Question	Marking guidance	Marks
04.2	- Higher temperatures melt land ice on Greenland which flows into the North Atlantic - The fresh water dilutes the seawater, reducing its salinity and density - Less dense water sinks less easily so the flow rate of the current is reduced - Less heat is transported to NW Europe so the UK could become colder (like Moscow / Churchill at the same latitude) even while global temperatures rise	4
04.3	- Several interconnected systems act at once over different timescales: jet stream changes may raise temperatures, a slowing Conveyor may lower them, greenhouse gases raise them - Natural fluctuations (solar output, orbit) can hide or exaggerate anthropogenic change; individual events do not show trends - Time delays between cause and effect (oceans take a long time to warm because of volume and heat capacity); incomplete understanding of the processes	3
04.4	Argo floats sink to a set depth for about 10 days, then surface and transmit temperature, salinity and position data; the sequence of positions shows current direction and speed (surface currents also by satellites and buoys)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 96-97; AQA spec 3.2.1.2, 3.2.2.3.	
05.1	- Location: discharge into the sea / large water bodies where currents dilute and disperse the pollutant - Location: chimneys / emissions sited downwind of urban areas so pollutants are not carried over populations; avoiding valleys and enclosed water bodies where pollutants concentrate - Location: not discharging wastes onto permeable rock above an aquifer / not siting landfill on aquifers - Timing: restrict polluting activities (industrial combustion, vehicle use) during temperature inversions when dispersal is poor - Timing: discharge liquid effluent when river flow is high / on an ebb tide so it is diluted and carried away - Timing: pesticide spraying at night / on still days; noisy operations restricted to daytime; night-flight restrictions	6
05.2	- The organisation / person that causes the pollution is responsible for (paying for) the damage caused - If the cost of preventing the pollution is less than the cost of the damage / fines, there is a financial incentive to prevent it; works best when penalties are high and escape is unlikely	2
05.3	- It is assumed that the waste will cause pollution / harm until research shows it is unlikely to - So release is not permitted until it has been tested; lack of knowledge of a problem is not an excuse / does not reduce responsibility	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	
06.1	B - rivers are accessible over a large area with few natural contaminants	1

Question	Marking guidance	Marks
06.2	• Potable / public supply: no pathogens, low toxins, good appearance / taste / smell (public health) • Industrial boiler / cooling water: soft water with low calcium so mineral deposits do not block pipes or reduce heat exchange; spray irrigation: low turbidity to avoid blocked pipes and low heavy metals	2
06.3	• Canals / pipes move water from areas with a surplus to areas with a shortage (eg Wales to England, China's Yangtze transfers) • Disadvantage: high cost / energy for pumping; ecological impacts on the donor river; transfer of pollutants or invasive species; political conflict	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-128; AQA spec 3.2.2.4-3.2.2.5.	
07.1	• Fission: splitting the nuclei of large atoms (U-235, Pu-239) when hit by neutrons • Fusion: joining the nuclei of small atoms (deuterium and tritium - isotopes of hydrogen) to form helium, releasing a neutron and energy - the energy source of stars	2
07.2	• Deuterium (hydrogen-2) extracted from (sea)water • Tritium (hydrogen-3) produced by neutron bombardment of lithium (a lithium blanket around the reactor)	2
07.3	• Very high temperature turns hydrogen into a plasma (electrons stripped away) and gives nuclei enough kinetic energy to overcome the repulsion between positive nuclei and collide • Vacuum so the plasma is not cooled by air • Magnetic field holds the plasma centrally so it does not touch the container walls and cool	3
07.4	• Fuel (deuterium / lithium) is abundant / effectively unlimited • Much less radioactive waste (no long-lived fission products); no chain reaction / meltdown risk; no weapons material • Not yet achieved sustained net energy output: reactors (JET) use more energy than they release; ITER aims for 500 MW out from 50 MW in; laser fusion also experimental; commercial viability decades away	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 195-196; AQA spec 3.3.4.1.	
08.1	• $(68\,000 - 7\,000) / 68\,000 \times 100$ • = 89.7% (accept 90%)	2
08.2	• Inflow from the rivers was reduced (sometimes to zero) so less water entered while evaporation continued • Evaporation removes pure water leaving the dissolved salts behind, so they became concentrated in a smaller volume	2
08.3	• Salinity rose above the range of tolerance of the freshwater fish species / osmotic stress; pollution by pesticides, fertilisers and industrial waste concentrated in the lake • Exposed lake bed of salt and pesticide-contaminated dust blown by wind onto farmland and settlements (health problems, soil salinisation) • Loss of wetland habitats and wildlife; local climate more extreme (loss of the moderating effect of the water body) • Loss of livelihoods; conflict between the countries sharing the rivers	4

Question	Marking guidance	Marks
08.4	- Upstream abstraction for irrigation reduces the flow available downstream - Upstream HEP dams hold water in summer for winter generation, reducing summer irrigation water downstream; growing populations increase demand	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117-118; AQA spec 3.2.2.1-3.2.2.2.	
09.1	- Vehicles: high energy density (200 Wh kg^{-1}) minimises battery mass so range is greater; many cycles and high efficiency - Stationary storage: mass is unimportant so the low energy density of lead-acid does not matter; lowest cost per kWh; well-established, recyclable - Lead-acid drawbacks: fewer cycles and lower efficiency; lithium fire risk needs management - Nickel-cadmium largely replaced because of cadmium toxicity and cost	4
09.2	<i>Any 2 from:</i> - Recharging speed - Self-discharge during non-use - Availability of raw materials (lithium, cobalt) - Recyclability / disposal	2
09.3	- Parked electric vehicles plugged into the grid form a very large distributed battery - At peaks a small proportion of their stored energy is fed back to the grid; surplus grid energy recharges them - avoiding costly, low-efficiency open-cycle gas turbines	2
09.4	- Surplus electricity electrolyses water to hydrogen, which can be converted to methane - Methane is fed into the existing natural gas pipe network and storage, so no new distribution system is needed	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 223-224; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - Organic nutrients (sewage, manure, silage effluent, paper mill / tannery / food processing effluent) are food for aerobic microbes whose respiration deoxygenates the water, killing fish and invertebrates; bacterial growths cover the bed, blocking light - Pathogens in sewage (cholera, typhoid, dysentery) spread via contaminated water; decay of organic matter releases inorganic nutrients - Inorganic nutrients: nitrates (fertiliser / manure runoff, leaching) and phosphates (sewage effluent, detergents, eroded soil) cause cultural eutrophication: algal blooms, shading of macrophytes, cyanobacterial toxins, night-time and end-of-summer deoxygenation - Human health: nitrites and methaemoglobinaemia; possible carcinogen via nitrosamines - Control of organic wastes: sewage treatment (pre-treatment, primary sedimentation removing >95% organic matter, aeration / activated sludge or trickling filters, anaerobic digestion of sludge with biogas) - Control of nutrients: tertiary phosphate stripping with iron(III) sulfate, denitrification in filter beds, micro-strainers / UV; Nitrate Vulnerable Zones, fertiliser timing / amount, low-solubility fertilisers, legumes, manure management; dredging sediments - Evaluation: sewage treatment has virtually eliminated deoxygenation and waterborne disease in developed countries but storm overflows and combined sewers still discharge untreated sewage; tertiary treatment is expensive so only used at sensitive sites; diffuse agricultural nitrate is hard to control and aquifers respond slowly; phosphate legacy in sediments - Judgement on point-source (sewage) control being more successful than diffuse-source (agriculture) control	9
10.2	Algae / cyanobacteria absorb nutrients directly from the water (not the sediments) so they use dissolved nutrients more easily and grow rapidly	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 266-272; AQA spec 3.4.3.2.9.	

Question	Marking guidance	Marks
11.1	Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Stages of investigation: identifying the issue (Halley Station UV data, Dobson unit measurements), hypothesis formation (Rowland-Molina 1974: persistence, UV dissociation, chlorine reactions), data collection (ground, satellite, balloon and aircraft air sampling detecting ClO), analysis of spatial, temporal and altitude variability, proposal of solutions, enactment • Why Antarctica: low stratospheric temperatures, ice crystals, stratospheric clouds, polar vortex, spring sunlight • Effects of UVB: skin cancer, cataracts, DNA damage, plant and marine damage - motivating action • Solutions: Vienna Convention, Montreal Protocol - phase-out and bans, alternative processes (pump sprays), alternative materials (HCFCs, HFCs, hydrocarbons, HFAs), collection and incineration of CFCs, fund for developing countries • Evaluation of success: ozone hole area / depth stabilising, CFC levels falling; limitations - persistence, remaining ODSs, HFCs as greenhouse gases, decades to full recovery • Comparison with other atmospheric problems (climate change, acid rain) - why ozone was easier; role of peer review, international cooperation, availability of substitutes • Conclusion on the extent to which it is a model for other issues Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-113; AQA spec 3.2.1.3.	25

Question	Marking guidance	Marks
11.2	Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Negative feedback: increased low-level cloud (albedo), increased photosynthesis and sequestration; hydrological cycle evaporation-precipitation • Positive feedback: melting permafrost (methane), reduced albedo of snow / ice / sea ice, methane hydrate release, ocean acidification and coral loss, forest and peat fires, soil decomposition, water vapour, cirrus clouds • Tipping points: self-sustaining processes; why control must precede them; examples • Implications: urgency and precautionary principle; need for deep cuts in greenhouse gases (energy, transport, buildings, agriculture, forestry, CCS) rather than only adaptation; geoengineering risks • Difficulty of prediction because feedbacks interact at different scales; models; international agreements and per capita footprints • Judgement on whether current strategies are adequate Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-107, 371-372; AQA spec 3.2.1.2, 3.6.1.	25